Sam Heller

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Education

Virginia Tech: College of Engineering, Blacksburg, VA *May 2027*
Bachelor of Science, Mechanical Engineering

Skills

- CAD: NX, SolidWorks, and Fusion 360
- Simulations: FEA and Stress Testing
- Coding: Python, HTML and MATLAB
- Rapid Prototyping: 3D Printing, CNC Machining
- Data Management: JMP, Excel, and Power BI
- AI Skills: Claude Code, Claude API and Agent Design
- Leadership: Hillel President and Hiking Club Founder

Experience

Mechanical Engineering Intern, Rhenus Automotive – Salem, VA *May 2026 – Aug 2026*
- Designed and built EasySDS, an automated chemical labeling compliance tool enforcing required GHS/OSHA fields, reducing per-label cost to under $1 and processing time by 90%.
- Reviewed and vetted 100+ work instructions for accuracy across the chassis and engine lines, editing and migrating documentation into the plant's Xeri system.
- Built a Power BI dashboard to help the safety team analyze incident data and automated approval notifications plant-wide using Power Automate.

Mechanical Design & Technical Animation Intern, DockBloxx – Atlanta, GA *May 2025 – Jan 2026*
- Modeled and animated the full product catalog of 20+ dock accessories, producing the CAD and assembly animations used across the company's website.
- Utilized CAD prototypes for feasibility analysis to ensure reliability and CNC manufacturability.
- Designed and developed a new modular product, now featured on the company's website, from concept through finalized prototype.

Operations Intern, Case-Mate – Atlanta, GA *May 2024 – Aug 2024*
- Analyzed shipping lane options across 11 carriers, by recommending route changes to lower delivery time 14%.
- Rebuilt inventory tracking for 300+ SKUs in NetSuite, saving the ops team 10+ hours per week.
- Built Excel models comparing carrier rates and lead times, creating a repeatable tool for quoting shipment costs.

Aerospace Intern, Applied Composites – San Diego, CA *May 2022 – Jun 2022*
- Designed a glider with a flying wing configuration using Fusion 360, Wing Helper, and Ansys.
- Built my design out of balsa wood and carbon fiber, then successfully completed flight tests.

Research

Undergraduate Research Assistant, NEEC Lab – Virginia Tech, Blacksburg, VA *Dec 2024 – May 2025*
- Led design and CAD development of biomimetic ears used in bat echolocation research experiments.
- Engineered ligament mechanisms enabling realistic ear motion and acoustic response.
- Applied 3D printing, silicone casting, and elastic material integration to create functional test models.

Projects

EasySDS, Chemical Labeling Compliance Web App | github.com/sam-j-heller/SDSLabelGenerator *May 2026 – Present*
- Built a Firebase-backed web app that generates OSHA HazCom 2012 and GHS compliant chemical labels, with facility-gated login, an approval queue, and multiple print formats.
- Architected the Firestore data model and admin tooling for per-facility chemical inventories, implementing with Claude Code and reviewing generated code against GHS field requirements.
- Validated output against BS 5609 certified vinyl label stock on a color laser print workflow and documented the process as a standard work instruction.

Microplastic-Filtering Kayak Paddle *Jul 2022 – Present*
- Designed and prototyped a kayak paddle that passively filters microplastics from the water column during a normal stroke, requiring no power or change to the user's technique.
- Integrated activated carbon filter media into the blade, sizing flow paths and media placement to capture particulate while keeping stroke resistance close to a standard paddle.
- Led a small design team from concept through functional prototype, iterating blade geometry with 3D printed test articles and on-water testing.
- Secured development advising from an international venture capital firm focused on sustainability.

Activities

President of Hillel at Virginia Tech *Nov 2024 – Dec 2025*
- Led a 400-student organization, overseeing all organizational initiatives and fostering a welcoming community.
- Restructured events to enhance participation and inclusivity, ensuring a diverse range of programming.